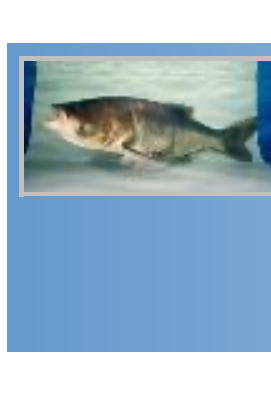
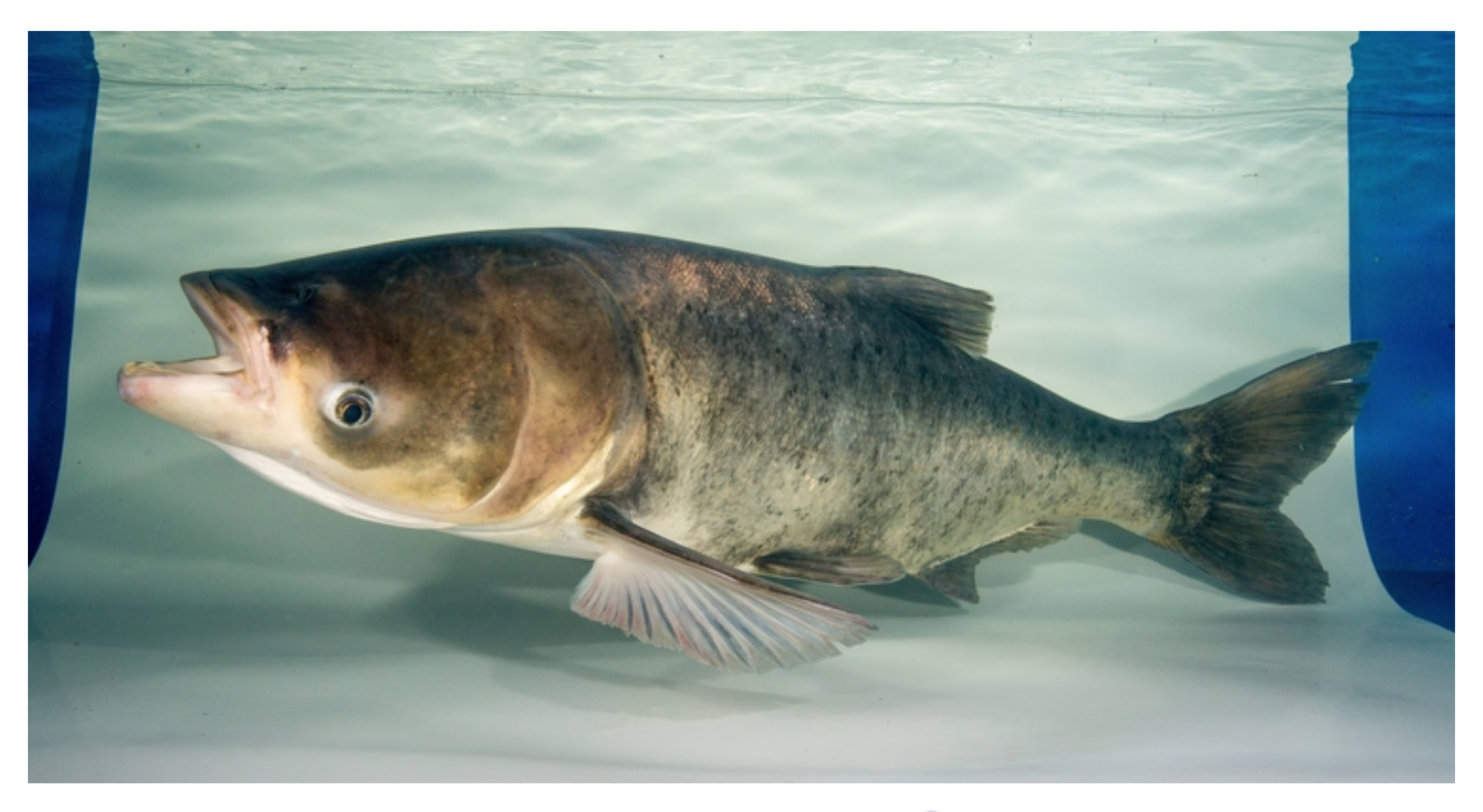




U.S. Fish and Wildlife Service

Hypophthalmichthys nobilis (Richardson, 1845)

Common name: Bighead Carp

Synonyms and Other Names: *Aristichthys nobilis* (Richardson, 1845), *Leuciscus nobilis* Richardson, 1845

Taxonomy: available through ITIS

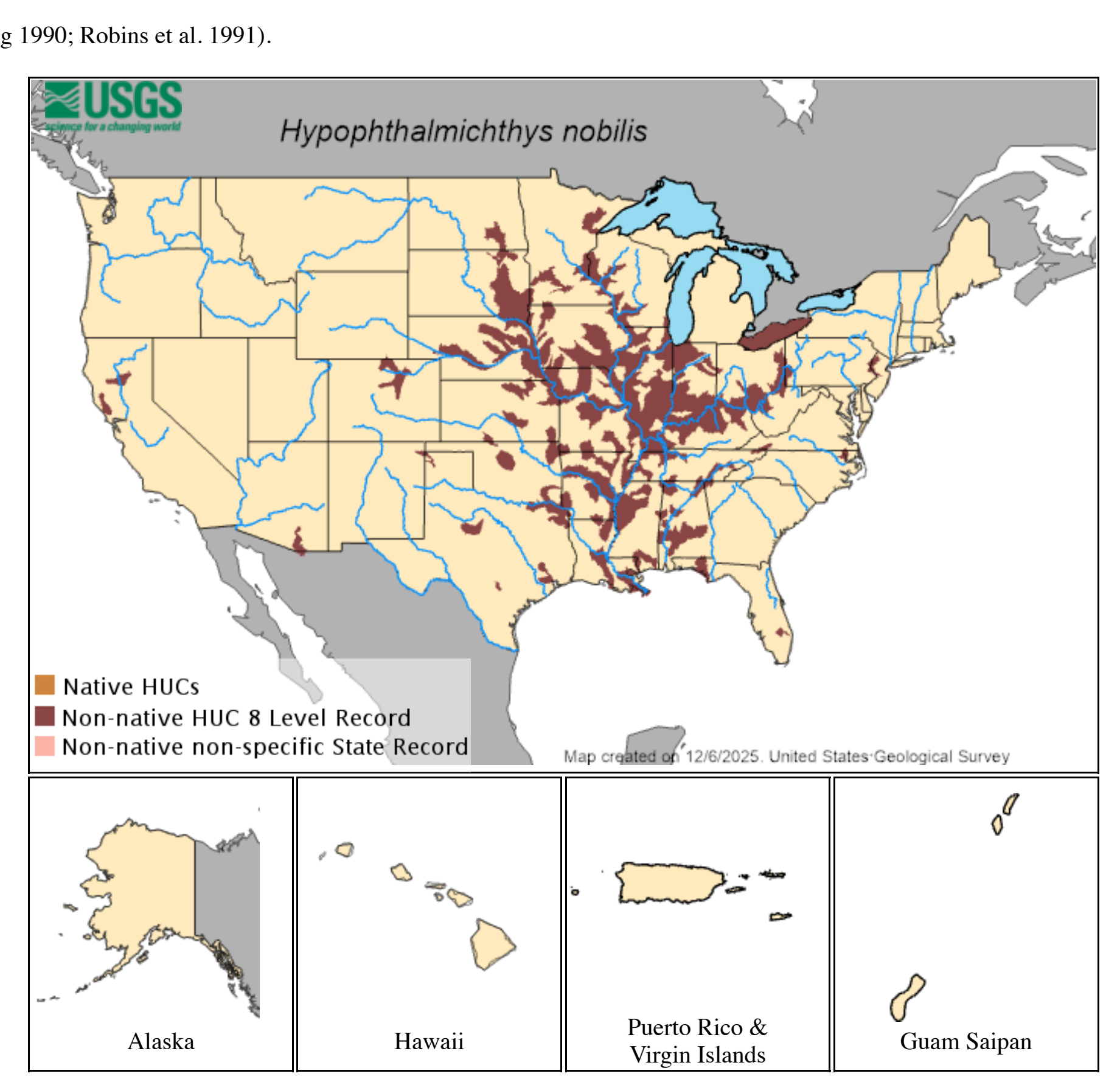
Injurious: This species is listed by the U.S. Fish and Wildlife Service as injurious wildlife.

Identification: The bighead carp is a large, narrow fish with eyes that project downward. Coloration of the body is dark gray, fading to white toward the underside, and with dark blotches on the sides. Its head has no scales, a large mouth with no teeth, and a protruding lower jaw. Its eyes are located far forward and low on its head. It is very similar to the silver carp, and can be distinguished by the dark coloration on its sides. The bighead carp can be identified by a smooth keel between the anal and pelvic fins that does not extend anterior of the base of the pelvic fins.

Distinguishing characteristics were given in Berg (1949) and Jennings (1988). Distinguishing characteristics, along with keys that include this species and photographs or illustrations also were included in a few of the more recently published state fish books (e.g., Robison and Buchanan 1988; Etnier and Starnes 1993; Pflieger 1997). A commonly used name is *Aristichthys nobilis*. Maximum size: 40 kg and 71.6 cm (Jennings 1988).

Size: 40 kg and 1.4 m

Native Range: Southern and central China (Li and Fang 1990; Robins et al. 1991).


USGS
science for a changing world
Hypophthalmichthys nobilis
Native HUCs
Non-native HUC 8 Level Record
Non-native non-specific State Record
Map created on 12/6/2025. United States Geological Survey
Alaska
Hawaii
Puerto Rico & Virgin Islands
Guam Saipan

Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Hypophthalmichthys nobilis* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AL	1984	2023	11	Bear; Black Warrior-Tombigbee; Guntersville Lake; Lower Alabama; Lower Tallapoosa; Middle Alabama; Middle Tombigbee-Chickasaw; Middle Tombigbee-Lubbub; Pickwick Lake; Upper Alabama; Upper Black Warrior
AZ	2007	2007	1	Upper Santa Cruz
AR	1988	2018	21	Bayou Macon; Bayou Meto; Boeuf; Cadron; Current; Dardanelle Reservoir; Frog-Mulberry; Lake Conway-Point Remove; Lower Arkansas; Lower Arkansas-Maumelle; Lower Black; Lower Little Arkansas, Oklahoma; Lower Mississippi-Greenville; Lower Mississippi-Memphis; Lower Ouachita-Bayou De Loutre; Lower St. Francis; Lower White; Lower White-Bayou Des Arc; Middle White; Upper Saline; Upper White-Village
CA	1992	2013	3	Big Chico Creek-Sacramento River; Russian; Thomes Creek-Sacramento River
CO	1980	2024	3	Cache La Poudre; Clear; Middle South Platte-Cherry Creek
FL	1989	2017	3	Lake Okeechobee; Lower Choctawhatchee; St. Andrew-St. Joseph Bays
IL	1986	2025	36	Bear-Wyaconda; Big Muddy; Cache; Cahokia-Joachim; Chicago; Copperas-Duck; Des Plaines; Embarras; Flint-Henderson; Highland-Pigeon; Iroquois; Kankakee; Little Calumet-Galien; Little Wabash; Lower Illinois; Lower Illinois; Lower Illinois-Lake Chautauqua; Lower Illinois-Senachwine Lake; Lower Kaskaskia; Lower Ohio; Lower Ohio-Bay; Lower Rock; Lower Sangamon; Lower Wabash; Macoupin; Middle Kaskaskia; Middle Wabash-Busseron; Peruque-Piasa; Pike-Root; Skillet; Spoon; The Sny; Upper Illinois; Upper Kaskaskia; Upper Mississippi-Cape Girardeau; Upper Sangamon
IN	1984	2025	18	Blue-Sinking; Eel; Highland-Pigeon; Little Calumet-Galien; Lower East Fork White; Lower Ohio-Little Pigeon; Lower Wabash; Lower White; Middle Ohio-Laughery; Middle Wabash-Busseron; Middle Wabash-Little Vermilion; Mississinewa; Muscatatuck; Ohio Region; Silver-Little Kentucky; Tippecanoe; Upper East Fork White; Upper Wabash
IA	1986	2024	19	Apple-Plum; Big Papillion-Mosquito; Blackbird-Soldier; Coon-Yellow; Copperas-Duck; Flint-Henderson; Keg-Weeping Water; Lake Red Rock; Little Sioux; Lower Big Sioux; Lower Des Moines; Lower Iowa; Middle Cedar; Middle Iowa; Nodaway; Platte; Skunk; Upper Chariton; West Nishnabotna
KS	1987	2023	10	Big Nemaha; Independence-Sugar; Little Arkansas; Lower Big Blue; Lower Kansas, Kansas; Lower Republican; Middle Neosho; Tarkio-Wolf; Upper Salt Fork Arkansas; Upper Walnut River
KY	1981	2024	21	Bayou De Chien-Mayfield; Blue-Sinking; Highland-Pigeon; Kentucky Lake; Licking; Little Scioto-Tygarts; Lower Cumberland; Lower Green; Lower Kentucky; Lower Mississippi-Memphis; Lower Ohio; Lower Ohio-Bay; Lower Ohio-Little Pigeon; Lower Tennessee; Middle Green; Middle Ohio-Laughery; Ohio Brush-Whiteoak; Rolling Fork; Rough; Salt; Silver-Little Kentucky
LA	1985	2024	13	Atchafalaya; Bayou Teche; Boeuf; East Central Louisiana Coastal; Lake Maurepas; Lake Pontchartrain; Lower Mississippi-Baton Rouge; Lower Mississippi-New Orleans; Lower Ouachita-Bayou De Loutre; Lower Pearl; Lower Red; Lower Red; Middle Red-Coushatta
MN	1996	2020	9	Buffalo-Whitewater; Coon-Yellow; Hawk-Yellow Medicine; La Crosse-Pine; Little Sioux; Lower St. Croix; Middle Minnesota; Rush-Vermillion; Twin Cities
MS	1986	2025	14	Big Sunflower; Black; Deer-Steele; Little Tallahatchie; Lower Mississippi-Greenville; Lower Mississippi-Helena; Lower Mississippi-Natchez; Lower Yazoo; Pascagoula; Pascagoula; Pickwick Lake; Tallahatchie; Upper Yazoo; Yalobusha
MO	1987	2022	20	Bear-Wyaconda; Bull Shoals Lake; Cahokia-Joachim; Independence-Sugar; Lake of the Ozarks; Lamine; Lower Chariton; Lower Grand; Lower Mississippi-Memphis; Lower Missouri; Lower Missouri-Crooked; Lower Missouri-Moreau; Lower Osage; One Hundred and Two; Peruque-Piasa; Tarkio-Wolf; The Sny; Town of Madrid-Saint Johns Bayou; Upper Mississippi-Cape Girardeau; Whitewater
NE	1990	2023	15	Big Papillion-Mosquito; Blackbird-Soldier; Keg-Weeping Water; Lewis and Clark Lake; Little Nemaha; Loup; Lower Elkhorn; Lower North Loup; Lower North Platte; Lower Platte; Middle Platte-Buffalo; Middle Platte-Prairie; Salt; Tarkio-Wolf; Upper Elkhorn
NJ	2010	2010	1	Middle Delaware-Musconetcong
NC	2011	2011	1	Chowan
ND	2019	2019	1	Upper James
OH	1995	2022	10	Hocking; Lake Erie; Little Muskingum-Middle Island; Lower Scioto; Mahoning; Middle Ohio-Laughery; Ohio Brush-Whiteoak; Raccoon-Symmes; Upper Ohio; Upper Ohio-Wheeling
OK	1992	2025	7	Bois D'arc-Island; Elk; Kiamichi; Lake O' The Cherokees; Lower Cimarron-Skeleton; Lower Neosho; Pecan-Waterhole
PA	2014	2014	1	Upper Ohio
SD	1998	2022	5	Lewis and Clark Lake; Lower Big Sioux; Lower James; Middle James; Vermillion
TN	1993	2017	9	Guntersville Lake; Kentucky Lake; Lower Cumberland; Lower Cumberland-Sycamore; Lower Hatchie; Lower Mississippi-Memphis; Middle Tennessee-Chickamauga; Obion; Watts Bar Lake
TX	1985	2015	9	Bois D'arc-Island; Buffalo-San Jacinto; Caddo Lake; Lake O'the Pines; Lower Sulpher; Rita Blanca; Upper Clear Fork Brazos; Upper San Antonio; West Fork San Jacinto
VA	1996	2007	1	South Fork Holston
WV	1997	2024	4	Lower Kanawha; Raccoon-Symmes; Upper Ohio-Shade; Upper Ohio-Wheeling
WI	1996	2020	9	Buffalo-Whitewater; Coon-Yellow; Grant-Little Maquoketa; La Crosse-Pine; Lower Chippewa; Lower St. Croix; Lower Wisconsin; Rush-Vermillion; St. Louis

Table last updated 12/8/2025

† Populations may not be currently present.

Ecology: Bighead carp is a powerful filter-feeder with a wide food spectrum that grows fast and reproduces quickly (Xie and Chen 2001), which makes this species a strong competitor. The diet of this species overlaps with that of planktivorous species (fish and invertebrates) and to some extent with that of the young of virtually all native fishes. Bighead carp are thought to deplete plankton stocks for native larval fishes and mussels (Laird and Page 1996). Bighead carp lack a true stomach which requires them to feed almost continuously (Henderson 1976).

Female bighead carp reach sexual maturity at three years of age, while males can reach sexual maturity in two years; however, this varies significantly with changing environmental conditions (Huet 1970; Kolar et al. 2007). Bigheaded carps are only known to spawn in large, turbulent rivers and it is believed that a rising hydrograph (flood event) is a primary spawning cue (Kolar et al. 2007). Fecundity increases with age and body weight and is directly related to growth rate (Verigin et al. 1990). In its native range, Bighead Carp has a fecundity ranging from 280,000-1.1 million eggs. In North America, fecundity ranged from 4,792-1.6 million eggs (Kipp et al. 2011). Bighead carp produce eggs that are semi-buoyant and require current to keep them from sinking to the bottom (Soin and Sukhanova 1972; Pflieger 1997). The eggs float for 40-60 hours before hatching.

Means of Introduction: Bighead carp were first imported into the United States in 1973 by a private fish farmer in Arkansas who wanted to use them in combination with other phytophagous fishes to improve water quality and increase fish production in culture ponds. In 1974 the Arkansas Game and Fish Commission and Auburn University, Alabama, obtained stock to assess their potential benefits and impacts (Jennings 1988). The species first began to appear in open waters, the Ohio and Mississippi rivers, in the early 1980s, likely as a result of escapes from aquaculture facilities (Jennings 1988). In April 1994, several thousand bighead carp, along with a few black carp Mylopharyngodon piceus, escaped into the Osage River, Missouri, when high water flooded hatchery ponds at an aquaculture facility near Lake of the Ozarks (Anonymous 1994). Fish that escaped into the Missouri River have increased and spread, since 1990, into the lower Kansas River of Kansas, and elsewhere (Cross and Collins 1995). The species may have dispersed into Oklahoma waters from fish illegally brought into southeast Kansas by a commercial fish farmer in 1988 (Pigg et al. 1993). According to Pigg et al. (1997), collections in the Grand River of Oklahoma since 1991 indicate a gradual downstream dispersal. The species was illegally stocked along with grass carp in one or a few ponds in California; these were brought into the state by a commercial aquaculturist. The live fish were reportedly transported in a concealed compartment under a load of black bass in the fall of 1989 from a fish grower in Oklahoma or Arkansas (Dill and Cordone 1997). The species was illegally stocked in Cherry Creek Reservoir, Colorado (P. Walker, personal communication).

Status: This species has been recorded from within, or along the borders of, at least 18 states. There is evidence of reproducing populations in the middle and lower Mississippi and Missouri rivers and the species is apparently firmly established in the states of Illinois and Missouri (Burr et al. 1996; Pflieger 1997). Pflieger (1997) received first evidence of natural reproduction, capture of young bighead carp, in Missouri in 1989. Burr and Warren (1993) reported on the taking of a postlarval fish in southern Illinois in 1992. Subsequently, Burr et al. (1996) noted that bighead carp appeared to be using the lower reaches of the Big Muddy, Cache, and Kaskaskia rivers in Illinois as spawning areas. Tucker et al. (1996) also found young-of-the-year in their 1992 and 1994 collections in the Mississippi River of Illinois and Missouri. Douglas et al. (1996) collected more than 1600 larvae of this genus from a backwater outlet of the Black River in Louisiana in 1994. The first open water record of this species in Arkansas is based on two specimens taken from the Arkansas River in 1986; however, as of the late 1980s there has been no evidence of natural reproduction in that state (Robison and Buchanan 1988). According to Dill and Cordone (1997), there is evidence that the California ponds containing Chinese carp have spilled since 1989, opening the door for bighead carp and grass carp to gain access to the Sacramento River. The West Virginia record involved a single fish taken in 1997 (Hoeft, personal communication). Harvest of bighead carp by commercial fishermen in Missouri has been somewhat erratic. In 1993, the species accounted for 0.6 percent (3,348 pounds) of the reported commercial fish harvest, a decline from the previous year (Robinson 1995).

Impact of Introduction:

Summary of species impacts derived from literature review. Click on an icon to find out more...

Ecological | **Economic** | **Human Health** | **Other**

The impact of this species in the United States is not adequately known. Because bighead carp are planktivorous and attain a large size, Laird and Page (1996) suggested these carp have the potential to deplete zooplankton populations. As Laird and Page pointed out, a decline in the availability of plankton can lead to reductions in populations of native species that rely on plankton for food, including all larval fishes, some adult fishes, and native mussels. Adult fishes most at risk from such competition in the Mississippi and Missouri rivers are paddlefish *Polyodon spathula*, bigmouth buffalo *Ictiobus cyprinellus*, and gizzard shad *Dorosoma petenense* (Burr et al. 1996; Pflieger 1997; Whitmore 1997; Tucker et al. 1998; Schrank et al. 2003). A study by Sampson et al. (2009) found that Invasive Carp (silver and bighead carps) had dietary overlap with gizzard shad and bigmouth buffalo, but not much of one with paddlefish.

Invasive carps have been shown to affect zooplankton communities (Burke et al. 1986, Lu et al. 2002, Cooke et al 2009; Calkins et al. 2012; Freedman et al. 2012; Sass et al. 2014).

Freedman et al. (2012) showed that resource use and trophic levels of the fish community change when Invasive Carps are present. They also demonstrated an impact on Bigmouth Buffalo and found isotopic values similar to Bluegill, Gizzard Shad, and Emerald Shiner.

Irons et al. (2007) showed significant declines in body condition of Gizzard Shad and Bigmouth Buffalo following invasion by Silver and Bighead carps. They state that ultimately, declines in body condition may decrease fecundity.

Remarks: Similar to the closely-related silver carp *Hypophthalmichthys molitrix*, the bighead carp is a filter feeder that prefers large river habitats. One of the so-called Chinese carps, it has been used in many parts of the world as a food fish and sometimes introduced in combination with silver carp into sewage lagoons and aquaculture ponds (Jennings 1988). In the United States bighead carp are frequently stocked into catfish culture ponds. According to Stickney (1996), studies have not confirmed that bighead carp actually do improve water quality in culture ponds.

Voucher specimens: Florida (UF 98162).

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Asian Carp Regionial Coordinating Committee

Proceedings of the Asian Carp Working Group Meeting, May 24, 2004, Columbia, Missouri

Facts About Bighead and Silver Carp (USGS)

Risk Assessment for Asian Carps in Canada (CSAS)

PA Sea Grant Fact Sheet

Global Invasive Species Database Factsheet

Michigan Asian Carp Management Plan

US Fish and Wildlife Service Ecological Risk Screening Summary for *Hypophthalmichthys nobilis*

FishBase Summary

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